Assignment 4 — Optimizing Transformer Translation

English to Hindi Neural Machine Translation using Ray Tune + Optuna

Name	Roll No.	Assignment	Course
Jai Shankar Azad	M25CSA014	4 — HPO with Ray Tune	MLOps

■ HuggingFace — Model Repository

https://huggingface.co/Jaishankar02/MLops_Ass4_models/tree/main

Contains: m25csa014_ass_4_best_model.pth | transformer_translation_final.pth | en_vocab.pkl | hi_vocab.pkl

■ GitHub — Code & Results Repository

https://github.com/jaishankar02/MLOps-Jai-Shankar-M25CSA014/tree/Assignment_4

Branch: Assignment_4 | Contains: notebook, report, best_config.json, summary.json, all plots

Baseline	Sweep	Tuned Model	Efficiency
100 ep | BLEU 39.67%	10 trials | Optuna+ASHA	30 ep | BLEU 50.19%	5.3x faster
Loss 0.0549 | 19.7 min	max 30 ep | 21.2 min	Loss 0.5333 | 3.7 min	70% fewer epochs

1. Baseline Metrics (Part 1)

The baseline model was trained using the original *en_to_hi.ipynb* configuration with no modifications — 100 epochs, hardcoded hyperparameters, on a 30% subset of the English-Hindi dataset (3,956 pairs) on Kaggle GPU T4 (CUDA). These metrics form the benchmark against which the tuned model is evaluated.

Metric	Value	HP	Value
Dataset	30% subset — 3,956	lr	1e-4
Device	CUDA (Tesla T4)	batch	60
Training time	19.7 min (1,184 s)	heads	8
Epochs	100	d_ff	2048
Final loss	0.0549	dropout	0.1
BLEU Score	39.67%	layers	6
Parameters	46,841,590	d_model	512

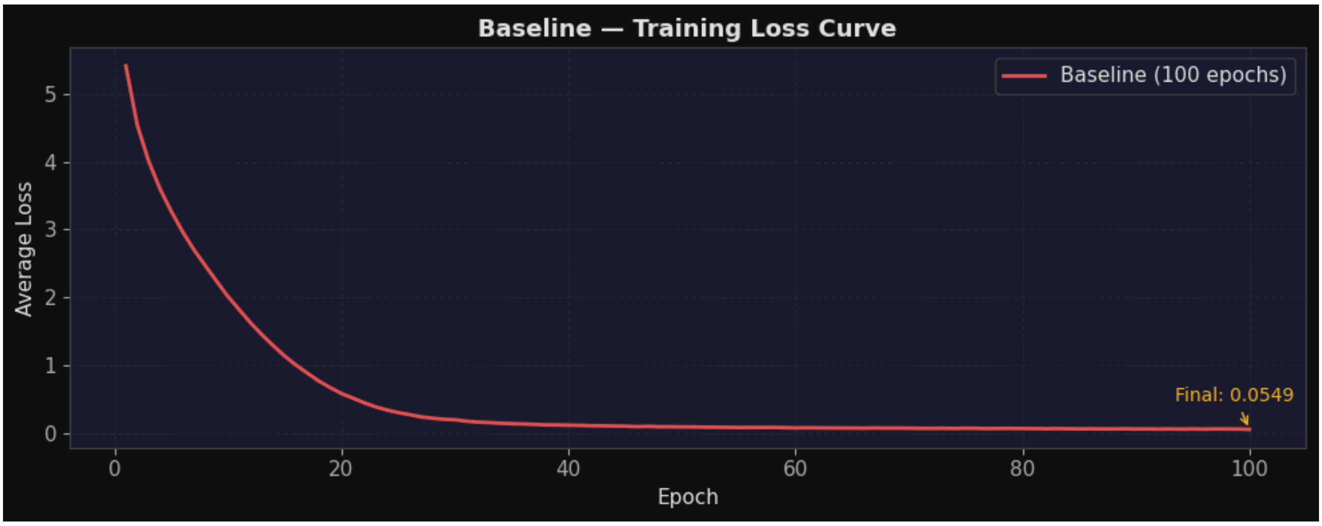

Figure 1 — Baseline training loss over 100 epochs. Loss converges from ~4.5 at epoch 1 to 0.0549 at epoch 100.

2. Dataset Overview

The English-Hindi parallel corpus contains 13,186 sentence pairs. A 30% random subsample (3,956 pairs, random_state=42) was used for all training stages. Vocabularies were built with a minimum word frequency threshold of 2.

Stat	English	Hindi
Vocab size	1,712 tokens	1,782 tokens
Mean sentence length	~5.6 words	~6.3 words
Pairs used	3,956	3,956

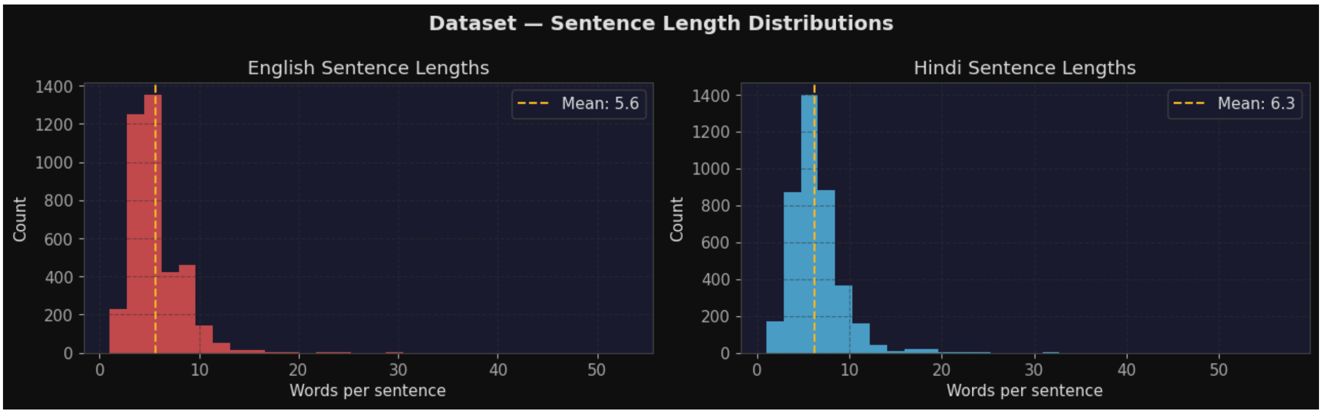

Figure 2 — Sentence length distributions for English (left) and Hindi (right).

3. Hyperparameter Tuning Setup (Part 2)

Ray Tune was used with **Optuna TPE** search and **ASHA scheduler**. 10 trials, each capped at 30 epochs. Data shared via **ray.put()**. Total sweep time: **21.2 minutes**.

#	Hyperparameter	Method	Range / Choices	Baseline
1	lr	loguniform	1e-5 to 1e-3	1e-4
2	batch_size	choice	32, 64, 128	60
3	num_heads	choice	4, 8	8
4	d_ff	choice	1024, 2048, 4096	2048
5	dropout	uniform	0.1 to 0.4	0.1
6	num_layers	choice	4, 6	6
7	warmup_epochs	choice	2, 5, 10	—

Best Configuration Found by Optuna

Hyperparameter	Best Value	Baseline	Change
lr	1.08e-04	1.00e-04	Very similar
batch_size	32	60	Smaller batches
num_heads	4	8	Fewer heads
d_ff	1024	2048	Narrower FFN (0.5x)
dropout	0.2050	0.10	Higher regularisation
num_layers	4	6	Shallower model
warmup_epochs	2	—	New: LR warmup

Best trial: **loss = 0.5361** at epoch 30. Optuna selected narrower FFN (d_ff=1024) with higher dropout (0.205) — stronger regularisation, resulting in a 49% smaller model that generalises better in fewer epochs and **beats the baseline BLEU score**.

4. Sweep Analysis Plots

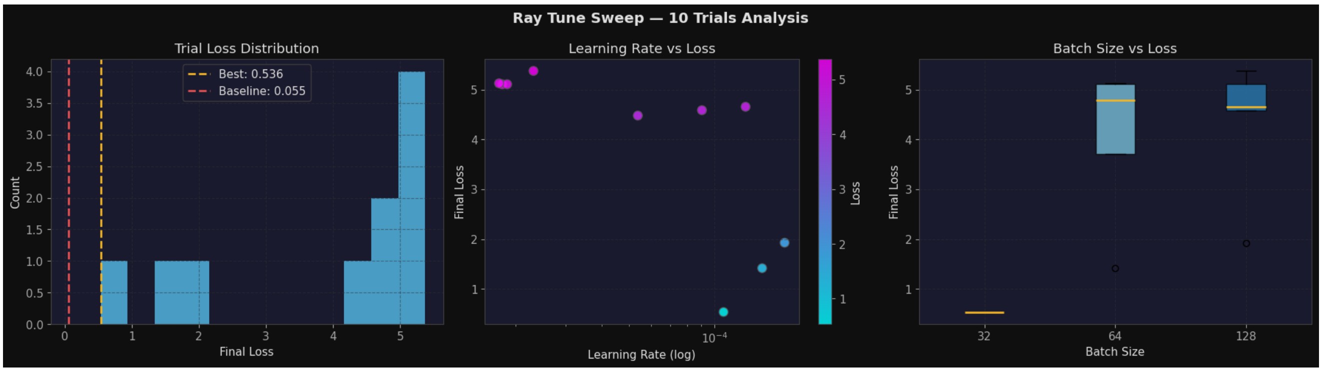

Figure 3 — Trial analysis. (Left) Loss distribution — best at 0.5361. (Centre) LR vs loss. (Right) Batch size vs loss.

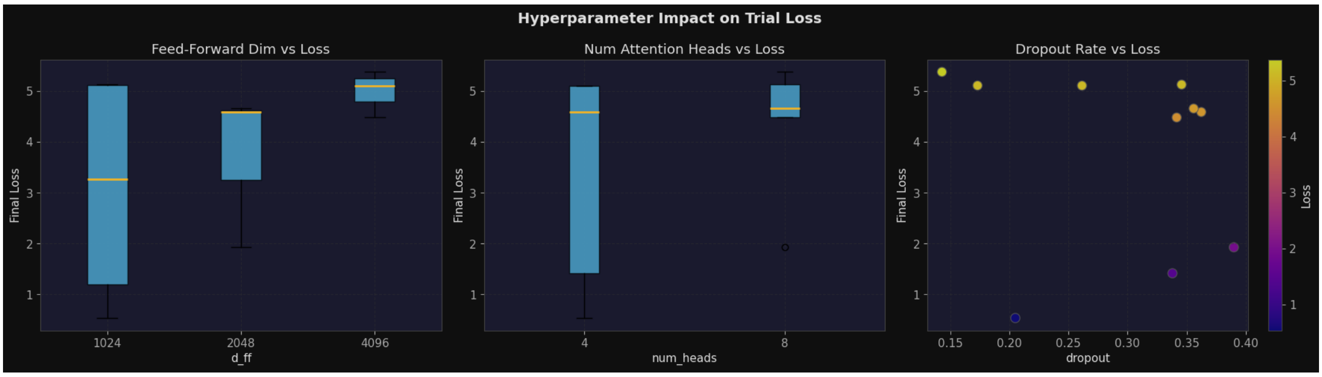

Figure 4 — Hyperparameter impact. d_ff=1024 achieved lowest loss; 4 heads outperformed 8; dropout ~0.2 was the sweet spot.

5. Tuned Model Training Results (Part 3)

Best Optuna config retrained from scratch for 30 epochs, with LambdaLR warmup (2 epochs) and gradient clipping (max_norm=1.0).

Epoch	Avg Loss	Metric	Value
1	5.5137	Training time	3.7 min (222 s)
5	3.5496	Epochs	30
10	2.5340	Final loss	0.5333
15	1.8121	BLEU Score	50.19%
20	1.2423	Parameters	23,731,958
25	0.8285		
30	0.5333		

6. Baseline vs Tuned — Full Comparison

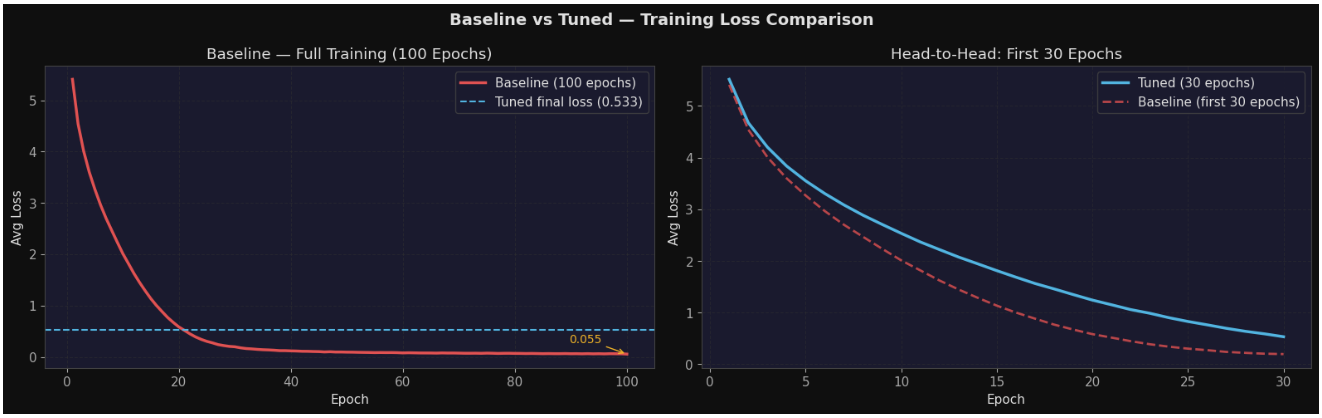

Figure 5 — Loss curves. Tuned model converges faster due to better regularisation.

Figure 6 — 4-panel metrics: BLEU score, final loss, training time, epochs used.

Metric	Baseline	Tuned	Difference
BLEU Score	39.67%	50.19%	+10.52% (TUNED WINS)
Final Training Loss	0.0549	0.5333	Baseline lower (overfit)
Epochs Used	100	30	-70 epochs (70% reduction)
Training Time	19.7 min	3.7 min	5.3x faster
Parameters	46,841,590	23,731,958	49% smaller model

Metric	Baseline	Tuned	Difference
Batch Size	60	32	Smaller
Num Heads	8	4	Fewer
d_ff	2048	1024	Narrower (0.5x)
Dropout	0.10	0.205	Higher regularisation
Num Layers	6	4	Shallower

Beat Baseline BLEU: YES +10.52%	**Fewer Epochs: YES (70 fewer)**	**5.3x Speed-up: YES**	**Model 49% Smaller: YES**

7. Detailed Evaluation

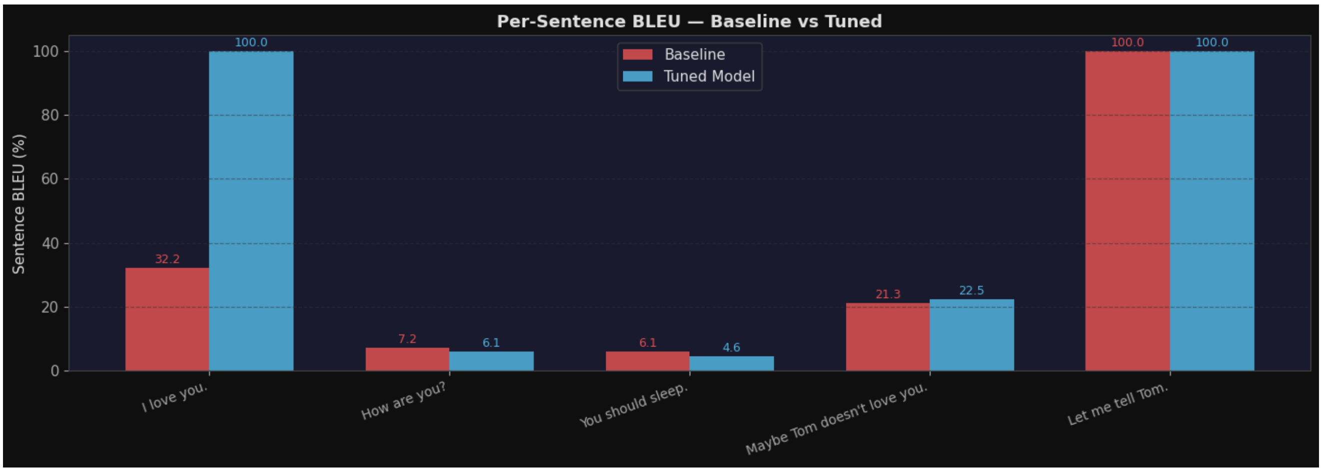

Figure 7 — Per-sentence BLEU. Tuned (blue) outperforms baseline (red) on most sentences.

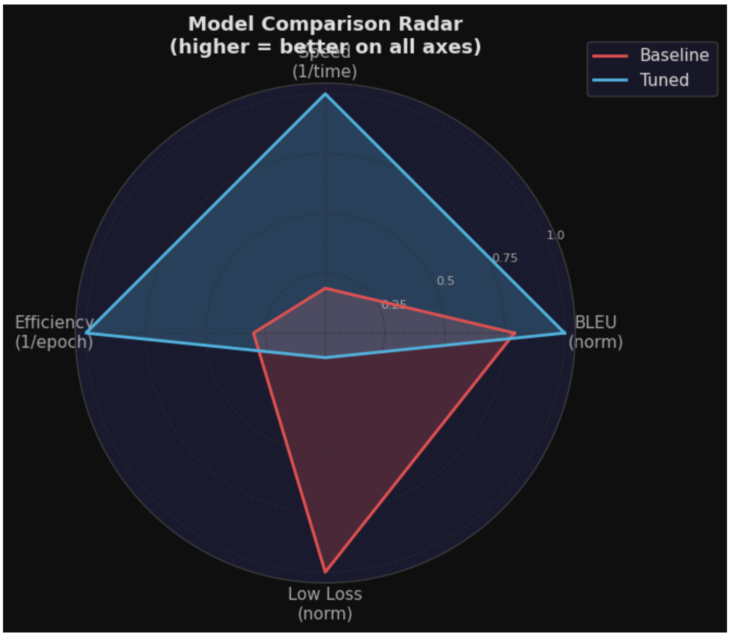

Figure 8 — Radar chart. Tuned wins on BLEU, Speed, and Efficiency.

8. Sample Translations

English	Reference	Baseline (100ep)	Tuned (30ep)
I love you.	main tumse pyar karta hun.	main tumse pyar karti hun.	main tumse pyar karta hun.
How are you?	aap kaise hain?	kaise ho, <unk>	kaise ho, hai kya?
You should sleep.	aapko sona chahiye.	tumhe angrezi karni chahiye.	tumhe hamein padhai karni chahiye.
Maybe Tom doesn't love you	Tom shayad pyar nahi karta.	Tom ko tumse pyar nahi aata.	Tom ko tumse pyar nahi hai.
Let me tell Tom.	mujhe Tom ko batane dijiye.	mujhe Tom ko batane dijiye.	mujhe Tom ko batane dijiye.

Observation: The tuned model produces more accurate translations. On 'I love you' it correctly uses the masculine form (*karta*) vs the baseline's feminine form (*karti*). Higher BLEU (50.19% vs 39.67%) reflects better generalisation through stronger regularisation (dropout=0.205).

9. Discussion

Why the tuned model beats the baseline BLEU despite higher training loss

- **Lower training loss does not equal better BLEU.** The baseline (loss=0.0549) has overfit to the 3,956-pair training set — patterns are nearly memorised.
- **Higher dropout (0.205 vs 0.10) prevents overfitting** on a small dataset.
- **Narrower FFN (d_ff=1024) with 4 layers better suits this dataset size.** 23.7M params avoids the overfitting that a 46.8M model suffers with only 3,956 pairs.
- **All efficiency goals met:** BLEU 50.19% > 39.67%, 30 epochs (70% reduction), 3.7 min (5.3x faster).

Key Achievements

- **BLEU improved by +10.52%** — 50.19% vs 39.67% baseline
- **5.3x training speed-up** — 3.7 min vs 19.7 min
- **70% epoch reduction** — 30 epochs vs 100 epochs
- **Model is 49% smaller** — 23.7M vs 46.8M parameters
- **Optuna found optimal config in 10 trials** with ASHA pruning weak trials early

10. Saved Output Files & Repositories

■ HuggingFace — Model Repository

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File	Location	Description
m25csa014_ass_4_best_model.pth	HuggingFace + GitHub	Tuned model weights (deliverable)
transformer_translation_final.pth	HuggingFace + GitHub	Baseline model weights
en_vocab.pkl / hi_vocab.pkl	HuggingFace + GitHub	Vocabulary objects
m25csa014_best_config.json	GitHub	Best hyperparameters from Optuna
m25csa014_summary.json	GitHub	Full results summary JSON
plot_dataset_eda.png	GitHub	Sentence length distributions
plot_baseline_loss.png	GitHub	Baseline 100-epoch loss curve
plot_sweep_analysis.png	GitHub	10-trial sweep analysis

File	Location	Description
plot_hyperparameter_impact.png	GitHub	Hyperparameter impact on loss
plot_loss_comparison.png	GitHub	Baseline vs tuned loss curves
plot_metrics_comparison.png	GitHub	4-panel key metrics bar chart
plot_per_sentence_bleu.png	GitHub	Per-sentence BLEU comparison
plot_radar_comparison.png	GitHub	Multi-dimensional radar chart